分组背包

人员

李瑞涵、田心一、柳力玮、赵书梵、刘宸熙、初锦阳、刘祺、明宗亮 到课, 蒋叔璋、苑钊 线上

上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/4431

#	用户名	姓名	编程分	时间	A (9 / 13)	B (6 / 7)	C (9 / 15)	D (9 / 11)	E (9 / 13)	F (5 / 6)	G (4 / 5)	H (4 / 5)	I (2 / 2)
1	zhaoshufan	赵书梵	9	10:42:58	0:12:14	0:15:42 ♥	(-1) 0:58:43	0:39:35 ♥	1:4:45 ♥	1:24:46 ♥	1:27:44 ♥	1:45:13 ♥	2:34:16 ♥
2	liuliwei	柳力玮	9	160:43:34	0:8:37 ♥	0:23:0	(-1) 0:56:51 ♥	0:46:54	1:27:6	1:30:16	1:38:20	76:43:12	76:49:18
3	tianxinyi	田心一	8	11:3:56	0:15:39	0:29:26	(-1) 1:16:42	0:51:16	1:37:9	1:43:32	(-1) 1:48:25	(-1) 2:1:47	
4	yangyongcheng	杨咏丞	8	406:55:47	0:59:29	147:36:19	1:25:33	0:53:26	1:44:29	1:51:36	126:9:24	126:15:31	
5	jiangshuzhang	蒋叔璋	5	6:11:4	0:14:41		(-1) 1:19:5	0:54:3	1:36:45	1:46:30			
6	jibohan	纪博涵	5	9:41:13	(-2) 1:20:44	2:1:11	(-2) 1:41:16	(-2) 0:50:13	1:47:49				
7	chujinyang	初锦阳	4	3:40:31	0:20:54		1:0:35	0:51:46	1:27:16				
8	liuchenxi	刘宸熙	4	5:2:20	(-2) 0:20:1	(-1)	1:31:24	0:48:57	1:41:58	(-1)			
9	liruihan	李瑞涵	4	5:42:7	0:14:43		1:28:41	0:49:36	(-4) 1:49:7				
10	liuqi	刘祺	1	167:21:25		167:21:25							

本周作业

https://cppoj.kids123code.com/contest/4524 (课上讲了 A ~ E 题, 课后作业是 E 题必做)

课堂表现

今天的第 5 题比较难, 同学们做起来难是正常的, 能把前 4 题掌握扎实就足够了。

课堂内容

通天之分组背包

f[i][j]: 考虑前 i 组物品, 当体积为 j 时, 最大价值是多少

```
#include <bits/stdc++.h>

using namespace std;

const int N = 1000 + 5, M = 100 + 5;
struct node {
  int c, v;
};
vector<node> vec[M];
```

```
int f[N];

int main()
{
  int n, m; cin >> m >> n;
  for (int i = 1; i <= n; ++i) {
    int c, v, id; cin >> c >> v >> id;
    vec[id].push_back({c,v});
  }

  for (int i = 1; i <= 100; ++i) {
    for (int j = m; j >= 0; --j) {
      for (node it : vec[i]) {
        if (j >= it.c) f[j] = max(f[j], f[j-it.c]+it.v);
      }
    }
  }
  cout << f[m] << endl;
  return 0;
}
```

AtCoder Janken 3

f[i][j]: 第 i 轮出 j 时, 最多赢多少场

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 2e5 + 5;
char s[maxn];
int w[maxn];
int f[maxn][3];

int iValue(char x) {
  if (x == 'R') return 0;
  if (x == 'P') return 1;
  return 2;
}

bool check(int a, int b) {
  if (a == 0 && b == 1) return true;
  if (a == 1 && b == 2) return true;
  if (a == 2 && b == 0) return true;
  return false;
}

int main()
{
  int n; cin >> n >> (s+1);
  for (int i = 1; i <= n; ++i) w[i] = iValue(s[i]);
```

```
  for (int i = 1; i <= n; ++i) {
    for (int j = 0; j < 3; ++j) {  // 第 i 个是 j
      if (check(j, w[i])) continue;
      for (int k = 0; k < 3; ++k) {  // 第 i-1 是 k
        if (k == j) continue;
        f[i][j] = max(f[i][j], f[i-1][k] + check(w[i], j));
      }
    }
  }

  cout << max({f[n][0], f[n][1], f[n][2]}) << endl;
  return 0;
}
```

Flip Cards

f[i][0/1]: 第 i 张是 正面/反面 朝上时, 有多少方案

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 2e5 + 5;
const int mod = 998244353;
int a[maxn], b[maxn], f[maxn][2];

signed main()
{
  int n; cin >> n;
  for (int i = 1; i <= n; ++i) cin >> a[i] >> b[i];

  f[1][0] = f[1][1] = 1;
  for (int i = 2; i <= n; ++i) {
    if (a[i] != a[i-1]) f[i][0] += f[i-1][0];
    if (a[i] != b[i-1]) f[i][0] += f[i-1][1];
    if (b[i] != a[i-1]) f[i][1] += f[i-1][0];
    if (b[i] != b[i-1]) f[i][1] += f[i-1][1];

    f[i][0] %= mod, f[i][1] %= mod;
  }

  cout << (f[n][0] + f[n][1]) % mod << endl;
  return 0;
}
```

Index × A(Not Continuous ver.)E

f[i][j]: 考虑前 i 个数, 选了 m 项时, 最大的和是多少

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 2000 + 5;
int w[maxn], f[maxn][maxn];

signed main()
{
  int n, m; cin >> n >> m;
  for (int i = 1; i <= n; ++i) cin >> w[i];

  memset(f, -0x3f, sizeof(f));
  for (int i = 0; i <= n; ++i) f[i][0] = 0;

  for (int i = 1; i <= n; ++i) {
    for (int j = 1; j <= m; ++j) {
      f[i][j] = max(f[i-1][j], f[i-1][j-1] + w[i]*j);
    }
  }
  cout << f[n][m] << endl;
  return 0;
}
```

[CSP-J 2022] 上升点列

f[i][j]: 考虑到第 i 项, 在中间添加了 j 个点时, 最长长度是多少

```
#include <bits/stdc++.h>

using namespace std;

const int N = 500 + 5, M = 100 + 5;
struct node {
  int x, y;
  bool operator < (const node& p) const {
    if (x != p.x) return x < p.x;
    return y < p.y;
  }
} w[N];
int f[N][M];

int main()
{
  int n, K; cin >> n >> K;
  for (int i = 1; i <= n; ++i) cin >> w[i].x >> w[i].y;
  sort(w+1, w+n+1);

  int res = 0;
```

```
  for (int i = 1; i <= n; ++i) {
    for (int j = 0; j <= K; ++j) {
      f[i][j] = j+1;
      for (int k = 1; k <= i-1; ++k) {
        if (w[i].x>=w[k].x && w[i].y>=w[k].y) {
          int delta = w[i].x-w[k].x + w[i].y-w[k].y-1;
          if (j-delta>=0) f[i][j] = max(f[i][j], f[k][j-delta]+delta+1);
        }
      }
    }

    res = max(res, f[i][K]);
  }
  cout << res << endl;
  return 0;
}
```